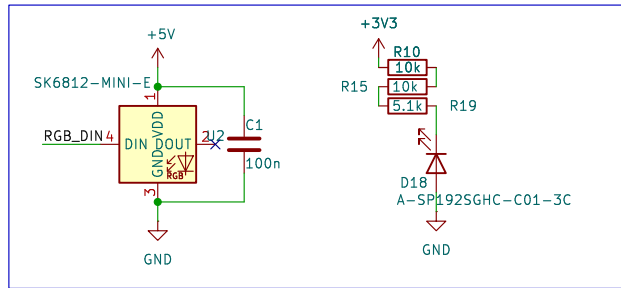
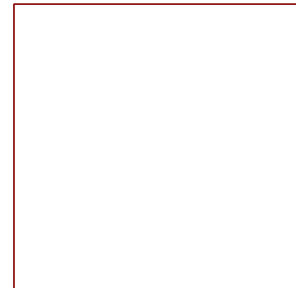


LEDs

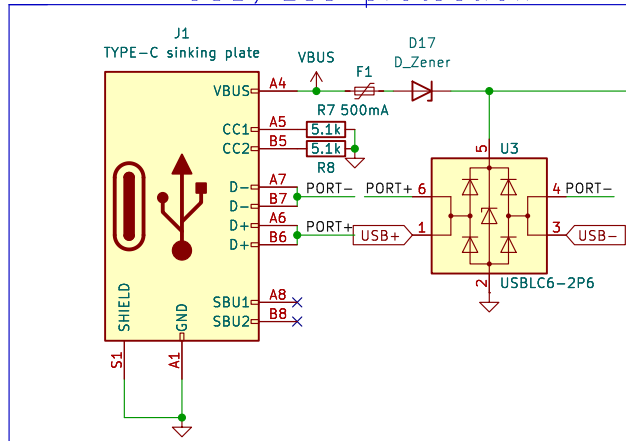


Switch Matrix

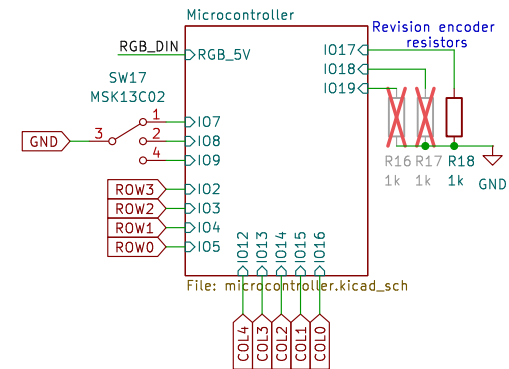
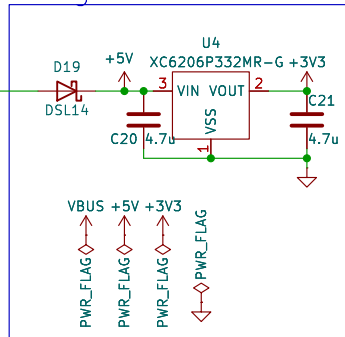


File: switch_matrix.kicad_sch

USB, ESD protection



Regulation



File: microcontroller.kicad_sch



Summacogni OÜ

Sheet: /
File: Mathpad.kicad_sch

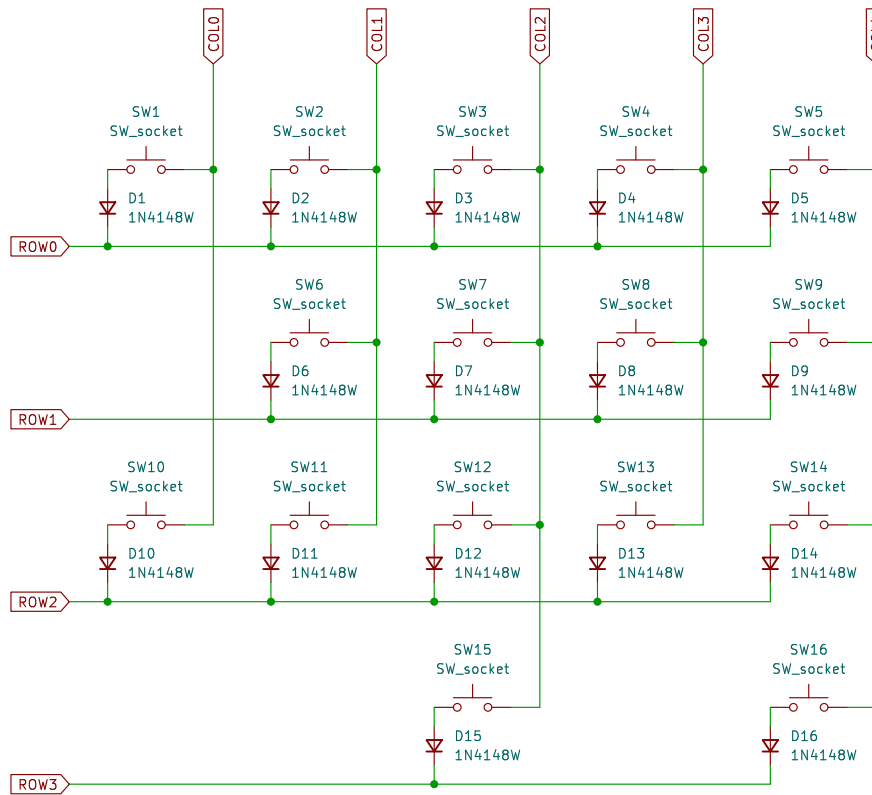
Title: Mathpad

Size: A4 Date: 2025-04-05

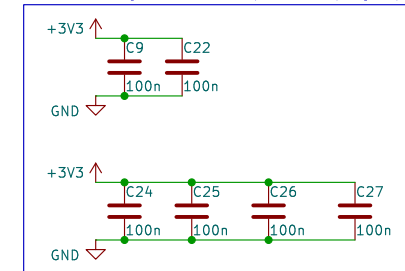
KiCad E.D.A. 8.0.4

Rev: 1.6

Id: 1/3



Matrix scanning current return path decoupling caps



Summacogni OÜ

Sheet: /Switch Matrix/
File: switch_matrix.kicad_sch

Title: Mathpad

Size: A4 Date: 2025-04-05
KiCad E.D.A. 8.0.4

Rev: 1.6
Id: 2/3

U1
W25Q64JV

3V3

C7
100n

QSPI_CS 1
QSPI_CLK 6
QSPI_SDO 5
QSPI_SD1 2
QSPI_SD2 3
QSPI_SD3 7

CS
CLK
DI(100)
DO(101)
IO2
IO3

VCC
GND
EP

The diagram shows a 3-state bus driver circuit. A RESET switch is connected to the gates of MOSFETs Q1A and Q2A through a 10k resistor R1. Q1A's source is grounded, and its drain is connected to the +3V3 supply through a 10k resistor R2. Q2A's source is grounded, and its drain is connected to the +3V3 supply through a 10k resistor R3. The gates of Q1A and Q2A are also connected to a common node that leads to the gate of Q2B through a 51k resistor R4. A delay circuit consisting of a 51k resistor R5 and two 4.7uF capacitors C3 and C4 is connected between the gates of Q2A and Q2B. The source of Q2B is grounded, and its drain is connected to the +3V3 supply through a 1k resistor R9. The output of the driver is the node between R9 and Q2B, which is connected to the QSPI_CS pin.

Id: 3/3